

B1 Fluids — Model Solutions, MT 2020

Question 1 — Oscillating plate (Stokes' second problem)

Problem. Incompressible viscous fluid above an infinite plate at $z = 0$ that oscillates as $\mathbf{u}|_{z=0} = (U \cos \omega t, 0)$; velocity decays for $z \rightarrow \infty$. Gravity neglected.

(a) Governing equations

Incompressible Navier–Stokes (no gravity):

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u}. \quad (1)$$

Mass conservation:

$$\nabla \cdot \mathbf{u} = 0.$$

Term-by-term: $\partial_t \mathbf{u}$ local acceleration; $(\mathbf{u} \cdot \nabla) \mathbf{u}$ advective (inertial) acceleration; $-\rho^{-1} \nabla p$ pressure-gradient force; $\nu \nabla^2 \mathbf{u}$ viscous diffusion of momentum; $\nabla \cdot \mathbf{u} = 0$ enforces incompressibility.

(b) Velocity field

Translation invariance: plate is infinite in x , forcing is x -independent:

$$\mathbf{u} = [u_x(z, t), u_z(z, t)], \quad p = p(z, t).$$

Mass conservation:

$$\frac{\partial u_z}{\partial z} = 0.$$

Apply $u_z \rightarrow 0$ as $z \rightarrow \infty$:

$$u_z \equiv 0.$$

Substitute into x -component of (1) (advection $u_z \partial_z u_x = 0$, no $\partial_x p$):

$$\frac{\partial u_x}{\partial t} = \nu \frac{\partial^2 u_x}{\partial z^2}. \quad (2)$$

Try $u_x = \text{Re}[u_0(z) e^{i\omega t}]$:

$$i\omega u_0 = \nu u_0''.$$

Rearrange:

$$u_0'' = \frac{i\omega}{\nu} u_0.$$

Characteristic roots $\pm k$ with $k^2 = i\omega/\nu$:

$$k = \sqrt{i\omega/\nu}.$$

Apply $\sqrt{i} = (1+i)/\sqrt{2}$:

$$k = (1+i)\sqrt{\omega/(2\nu)}.$$

Define $\delta = \sqrt{2\nu/\omega}$:

$$k = \frac{1+i}{\delta}.$$

Decay as $z \rightarrow \infty$ kills e^{+kz} :

$$u_0(z) = Ue^{-kz}.$$

Apply $u_x(0, t) = U \cos \omega t$, fixing the amplitude U real:

$$u_x(z, t) = \text{Re} \left[U e^{-(1+i)z/\delta} e^{i\omega t} \right].$$

Combine exponentials in the bracket:

$$u_x(z, t) = \text{Re} \left[U e^{-z/\delta} e^{i(\omega t - z/\delta)} \right].$$

Take the real part:

$$\boxed{u_x(z, t) = U e^{-z/\delta} \cos\left(\omega t - \frac{z}{\delta}\right), \quad u_z = 0} \quad (3)$$

(c) Stress tensor

Newtonian stress:

$$\sigma_{ij} = -p \delta_{ij} + \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right).$$

Use $u_z = 0$ and $\partial_x u_x = 0$:

$$\sigma_{xx} = -p, \quad \sigma_{zz} = -p.$$

Off-diagonal:

$$\sigma_{xz} = \sigma_{zx} = \mu \frac{\partial u_x}{\partial z}.$$

Differentiate the exponential prefactor:

$$\frac{\partial}{\partial z} \left[U e^{-z/\delta} \right] = -\frac{U}{\delta} e^{-z/\delta}.$$

Differentiate the phase using $\partial_z \cos(\omega t - z/\delta) = (1/\delta) \sin(\omega t - z/\delta)$:

$$\frac{\partial u_x}{\partial z} = -\frac{U}{\delta} e^{-z/\delta} \cos\left(\omega t - \frac{z}{\delta}\right) + \frac{U}{\delta} e^{-z/\delta} \sin\left(\omega t - \frac{z}{\delta}\right).$$

Factor $-U e^{-z/\delta}/\delta$:

$$\frac{\partial u_x}{\partial z} = -\frac{U}{\delta} e^{-z/\delta} \left[\cos\left(\omega t - \frac{z}{\delta}\right) - \sin\left(\omega t - \frac{z}{\delta}\right) \right].$$

Apply $\cos \alpha - \sin \alpha = \sqrt{2} \cos(\alpha + \pi/4)$:

$$\frac{\partial u_x}{\partial z} = -\frac{\sqrt{2} U}{\delta} e^{-z/\delta} \cos\left(\omega t - \frac{z}{\delta} + \frac{\pi}{4}\right).$$

Therefore

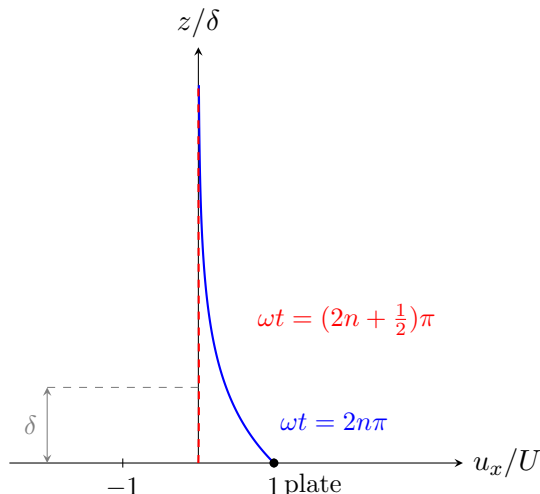
$$\boxed{\sigma_{xz} = \sigma_{zx} = -\mu \frac{\sqrt{2} U}{\delta} e^{-z/\delta} \cos\left(\omega t - \frac{z}{\delta} + \frac{\pi}{4}\right), \quad \sigma_{xx} = \sigma_{zz} = -p}$$

(p undetermined to within an additive constant since $\partial_z p = 0$ and gravity is neglected).

(d) Profiles, force balance, boundary layer

Profiles:

$$u_x(z, t = 2n\pi/\omega) = Ue^{-z/\delta} \cos(z/\delta), \quad u_x(z, t = (2n + \frac{1}{2})\pi/\omega) = Ue^{-z/\delta} \sin(z/\delta).$$



Force balance: in (2) unsteady inertia $\partial_t u_x$ balances viscous diffusion $\nu \partial_z^2 u_x$. Pressure is uniform; advection vanishes. Momentum injected at the plate diffuses upward while the imposed phase reverses every half period, generating the damped travelling wave (3).

Boundary-layer thickness from $\delta = \sqrt{2\nu/\omega}$:

$$\delta = \sqrt{2\nu/\omega}$$

Independent of U . Physical reading: in one forcing period $T = 2\pi/\omega$ vorticity diffuses a distance $\sqrt{\nu T} \sim \delta$ before the surface reverses, halting further penetration.

Tidal-flow application: by Galilean change of frame to one in which the fluid oscillates at $-U \cos \omega t$ and the plate is fixed, the same equation governs flow over a stationary bed with oscillatory far-field velocity. So an oscillatory Stokes layer of thickness $\delta = \sqrt{2\nu/\omega}$ does form near a tidal bed. Numerical estimate ($\nu = 10^{-6} \text{ m}^2 \text{ s}^{-1}$, $\omega = 1.4 \times 10^{-4} \text{ s}^{-1}$ semi-diurnal): $\delta \sim 10 \text{ cm}$. Caveats: real tidal flow is turbulent (large Re), so the laminar Stokes layer is replaced by a thicker turbulent boundary layer; the laminar solution describes only the viscous sub-layer.

Question 2 — Active cylinder, potential flow

Problem. 2D inviscid, incompressible, irrotational flow past cylinder $r = a$ with prescribed normal velocity $u_r(a, \theta) = A \sin \theta$, far-field $\mathbf{u} \rightarrow -U \hat{\mathbf{x}}$, $p \rightarrow p_0$. Cylinder axis vertical.

(a) Velocity potential and Bernoulli

Irrotational: $\nabla \times \mathbf{u} = 0$, so $\mathbf{u} = \nabla \phi$. Incompressible: $\nabla \cdot \mathbf{u} = 0$, hence

$$\nabla^2 \phi = 0.$$

Euler with gravity:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p - g \hat{\mathbf{z}}.$$

Vector identity:

$$(\mathbf{u} \cdot \nabla) \mathbf{u} = \nabla(\tfrac{1}{2}|\mathbf{u}|^2) - \mathbf{u} \times (\nabla \times \mathbf{u}).$$

Apply $\nabla \times \mathbf{u} = 0$:

$$(\mathbf{u} \cdot \nabla) \mathbf{u} = \nabla(\tfrac{1}{2}|\mathbf{u}|^2).$$

Apply $\mathbf{u} = \nabla\phi$ in the local-acceleration term:

$$\frac{\partial \mathbf{u}}{\partial t} = \frac{\partial}{\partial t} \nabla\phi = \nabla\left(\frac{\partial\phi}{\partial t}\right).$$

Write $-g\hat{\mathbf{z}} = -\nabla(gz)$ and $-\nabla p/\rho = -\nabla(p/\rho)$ (constant ρ):

$$\nabla\left(\frac{\partial\phi}{\partial t} + \tfrac{1}{2}|\nabla\phi|^2 + \frac{p}{\rho} + gz\right) = 0.$$

Therefore $\nabla H = 0$ with

$$\boxed{H = \frac{\partial\phi}{\partial t} + \tfrac{1}{2}|\mathbf{u}|^2 + \frac{p}{\rho} + gz} \quad (4)$$

Steady interpretation: $\partial_t\phi = 0$, so H is the conserved total mechanical energy per unit mass — kinetic + pressure-work term + gravitational potential.

(b) Solution and surface pressure

BCs:

$$\left.\frac{\partial\phi}{\partial r}\right|_{r=a} = A \sin\theta, \quad \nabla\phi \rightarrow -U\hat{\mathbf{x}} \text{ as } r \rightarrow \infty.$$

Resolve $-U\hat{\mathbf{x}} = -U \cos\theta \hat{\mathbf{r}} + U \sin\theta \hat{\boldsymbol{\theta}}$:

$$\phi \rightarrow -Ur \cos\theta \quad (r \rightarrow \infty).$$

From the general expansion keep $n = 1$ modes only (BCs only excite $\sin\theta, \cos\theta$):

$$\phi = \left(A_1 r + \frac{B_1}{r}\right) \cos\theta + \left(C_1 r + \frac{D_1}{r}\right) \sin\theta.$$

Take $r \rightarrow \infty$, where $B_1/r, D_1/r \rightarrow 0$:

$$\phi \rightarrow A_1 r \cos\theta + C_1 r \sin\theta.$$

Match to $-Ur \cos\theta$:

$$A_1 = -U, \quad C_1 = 0.$$

Differentiate ϕ in r :

$$\frac{\partial\phi}{\partial r} = \left(A_1 - \frac{B_1}{r^2}\right) \cos\theta + \left(C_1 - \frac{D_1}{r^2}\right) \sin\theta.$$

Evaluate at $r = a$ with $A_1 = -U, C_1 = 0$:

$$\left(-U - \frac{B_1}{a^2}\right) \cos\theta + \left(-\frac{D_1}{a^2}\right) \sin\theta = A \sin\theta.$$

Match coefficients:

$$B_1 = -Ua^2, \quad D_1 = -Aa^2.$$

Hence

$$\boxed{\phi(r, \theta) = -U \left(r + \frac{a^2}{r} \right) \cos \theta - \frac{Aa^2}{r} \sin \theta} \quad (5)$$

Velocity:

$$u_r = \frac{\partial \phi}{\partial r} = -U \left(1 - \frac{a^2}{r^2} \right) \cos \theta + \frac{Aa^2}{r^2} \sin \theta,$$

$$u_\theta = \frac{1}{r} \frac{\partial \phi}{\partial \theta} = U \left(1 + \frac{a^2}{r^2} \right) \sin \theta - \frac{Aa^2}{r^2} \cos \theta.$$

At $r = a$:

$$u_r(a, \theta) = A \sin \theta, \quad u_\theta(a, \theta) = 2U \sin \theta - A \cos \theta.$$

Surface speed squared:

$$|\mathbf{u}(a, \theta)|^2 = A^2 \sin^2 \theta + (2U \sin \theta - A \cos \theta)^2.$$

Expand the square:

$$(2U \sin \theta - A \cos \theta)^2 = 4U^2 \sin^2 \theta - 4UA \sin \theta \cos \theta + A^2 \cos^2 \theta.$$

Insert into $|\mathbf{u}|^2$:

$$|\mathbf{u}|^2 = A^2 \sin^2 \theta + 4U^2 \sin^2 \theta - 4UA \sin \theta \cos \theta + A^2 \cos^2 \theta.$$

Apply $\sin^2 \theta + \cos^2 \theta = 1$ to combine the A^2 terms:

$$|\mathbf{u}|^2 = A^2 + 4U^2 \sin^2 \theta - 4UA \sin \theta \cos \theta.$$

Apply $2 \sin \theta \cos \theta = \sin 2\theta$:

$$|\mathbf{u}|^2 = A^2 + 4U^2 \sin^2 \theta - 2UA \sin 2\theta.$$

Apply steady Bernoulli between far-field and the cylinder surface (same horizontal level, gz cancels):

$$p_0 + \frac{1}{2} \rho U^2 = p(a, \theta) + \frac{1}{2} \rho |\mathbf{u}(a, \theta)|^2.$$

Solve for $p(a, \theta)$:

$$p(a, \theta) = p_0 + \frac{1}{2} \rho U^2 - \frac{1}{2} \rho [A^2 + 4U^2 \sin^2 \theta - 2UA \sin 2\theta].$$

Apply $\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$ to $4U^2 \sin^2 \theta$:

$$4U^2 \sin^2 \theta = 2U^2 - 2U^2 \cos 2\theta.$$

Substitute into $-\frac{1}{2} \rho [A^2 + 4U^2 \sin^2 \theta - 2UA \sin 2\theta]$:

$$-\frac{1}{2} \rho [\dots] = -\frac{1}{2} \rho A^2 - \rho U^2 + \rho U^2 \cos 2\theta + \rho U A \sin 2\theta.$$

Add the $p_0 + \frac{1}{2} \rho U^2$ outside:

$$p(a, \theta) = p_0 + \frac{1}{2} \rho U^2 - \frac{1}{2} \rho A^2 - \rho U^2 + \rho U^2 \cos 2\theta + \rho U A \sin 2\theta.$$

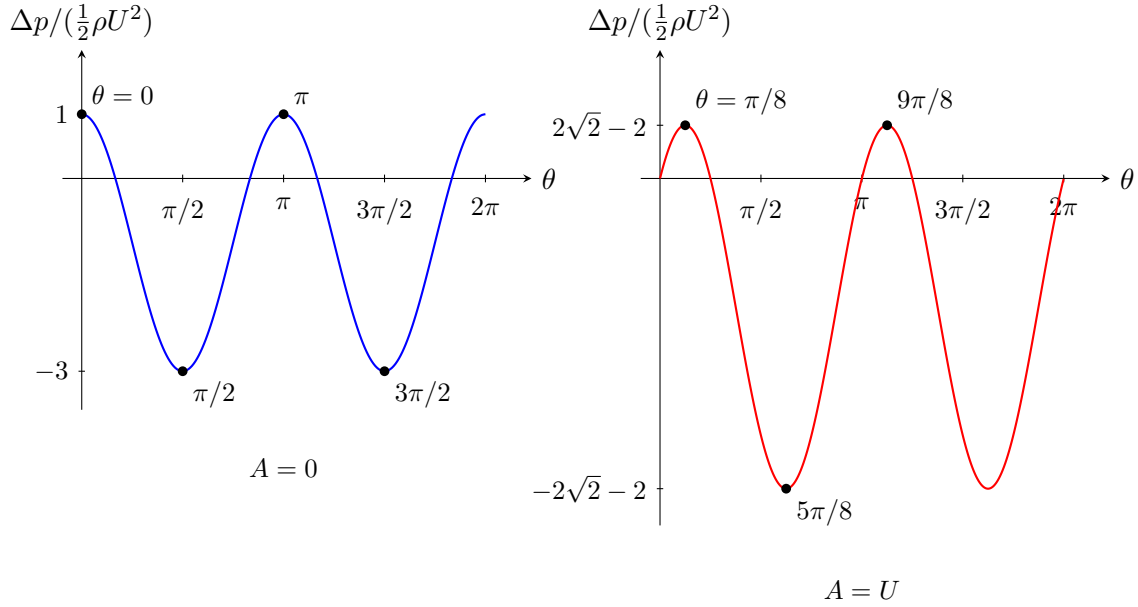
Combine $\frac{1}{2} \rho U^2 - \rho U^2 = -\frac{1}{2} \rho U^2$:

$$p(a, \theta) = p_0 - \frac{1}{2} \rho U^2 - \frac{1}{2} \rho A^2 + \rho U^2 \cos 2\theta + \rho U A \sin 2\theta.$$

Factor $\frac{1}{2} \rho U^2$:

$$\boxed{p(a, \theta) = p_0 + \frac{1}{2} \rho U^2 \left[2 \cos 2\theta - 1 - \frac{A^2}{U^2} + 2 \frac{A}{U} \sin 2\theta \right]} \quad (6)$$

Sketch of $\Delta p \equiv p(a, \theta) - p_0$ in units of $\frac{1}{2} \rho U^2$:



For $A = 0$, fore-aft symmetric about $\theta = \pi/2, 3\pi/2$; for $A = U$, the curve shifts and tilts: maximum suction at $\theta = 5\pi/8$, peak pressure at $\theta = \pi/8$ — a transverse asymmetry that lifts the cylinder.

(c) Boundary-layer separation

Inert case $A = 0$. Inviscid surface speed $u_\theta = 2U \sin \theta$, so a no-slip boundary layer grows everywhere except the stagnation points $\theta = 0, \pi$. Surface pressure gradient (using (6) with $A = 0$):

$$\frac{1}{a} \frac{\partial p}{\partial \theta} = -\frac{2\rho U^2}{a} \sin 2\theta.$$

Adverse on the rear half, $\pi/2 < \theta < \pi$ (and mirror $\pi < \theta < 3\pi/2$). Separation occurs shortly after $\theta = \pi/2$ on top and shortly after $\theta = 3\pi/2$ on the bottom (rear shoulders).

Active case $A = U$, upper surface $0 < \theta < \pi$. Differentiate (6):

$$\frac{1}{a} \frac{\partial p}{\partial \theta} = \frac{\rho U^2}{a} [-2 \sin 2\theta + 2 \cos 2\theta].$$

Locate the surface pressure extrema by $\partial_\theta p = 0$:

$$\tan 2\theta = 1 \Rightarrow \theta = \pi/8 \text{ (maximum)}, \theta = 5\pi/8 \text{ (minimum)}.$$

Inviscid surface tangential velocity $u_\theta(a, \theta) = 2U \sin \theta - U \cos \theta$, vanishing at the shifted stagnation $\tan \theta = 1/2$ ($\theta \approx 0.46$). Downstream of that stagnation on $\theta > 0$, flow runs in $+\theta$ and pressure minimum is reached at $\theta = 5\pi/8$; adverse region is $5\pi/8 < \theta < \pi$. Compared with $A = 0$ (adverse from $\theta = \pi/2$) the adverse-pressure region starts later, pushing the would-be separation point further back. The imposed $u_r = A \sin \theta > 0$ also blows fluid *outward* from the wall on $\theta > 0$, energising the boundary layer (analogous to suction-blowing control). Both effects *inhibit* separation on the upper surface relative to $A = 0$.

Question 3 — Vorticity dynamics and shear-layer mode

Problem. Vorticity equation, piecewise-shear base flow, linear interfacial mode.

(a) Vorticity equation

Take curl of incompressible Navier–Stokes:

$$\frac{\partial \boldsymbol{\omega}}{\partial t} + \nabla \times [(\mathbf{u} \cdot \nabla) \mathbf{u}] = \nu \nabla^2 \boldsymbol{\omega}.$$

Identity:

$$(\mathbf{u} \cdot \nabla) \mathbf{u} = \nabla(\tfrac{1}{2}|\mathbf{u}|^2) - \mathbf{u} \times \boldsymbol{\omega}.$$

Take curl, use $\nabla \times \nabla(\cdot) = 0$ and $\nabla \cdot \mathbf{u} = \nabla \cdot \boldsymbol{\omega} = 0$:

$$\nabla \times (\mathbf{u} \times \boldsymbol{\omega}) = (\boldsymbol{\omega} \cdot \nabla) \mathbf{u} - (\mathbf{u} \cdot \nabla) \boldsymbol{\omega}.$$

Combine:

$$\boxed{\frac{D\boldsymbol{\omega}}{Dt} = (\boldsymbol{\omega} \cdot \nabla) \mathbf{u} + \nu \nabla^2 \boldsymbol{\omega}} \quad (7)$$

Term-by-term: $D\boldsymbol{\omega}/Dt$ material rate of vorticity change following a fluid parcel; $(\boldsymbol{\omega} \cdot \nabla) \mathbf{u}$ vortex stretching/tilting by the velocity gradient; $\nu \nabla^2 \boldsymbol{\omega}$ viscous diffusion of vorticity.

2D inviscid: $\boldsymbol{\omega} = \omega_z(x, y, t) \hat{\mathbf{z}}$, no z -gradients in $\mathbf{u}$:

$$(\boldsymbol{\omega} \cdot \nabla) \mathbf{u} = \omega_z \partial_z \mathbf{u} = 0.$$

Drop viscosity:

$$\frac{D\omega_z}{Dt} = 0.$$

Streamfunction: $u_x = \partial_y \psi$, $u_y = -\partial_x \psi$, so $\omega_z = \partial_x u_y - \partial_y u_x = -\nabla^2 \psi$:

$$\frac{D}{Dt} \nabla^2 \psi = 0. \quad (8)$$

(b) Base flow

From $\bar{U} = \partial_y \bar{\psi}$:

$$\bar{\psi}(y) = \int_0^y \bar{U}(y') dy'.$$

Piecewise integration:

$$\bar{\psi}(y) = \begin{cases} -\frac{1}{2}\alpha y^2, & y \geq 0, \\ 0, & y < 0. \end{cases}$$

Vorticity $\bar{\omega}_z = -\partial_y^2 \bar{\psi}$:

$$\boxed{\bar{\omega}_z(y) = \begin{cases} \alpha, & y > 0, \\ 0, & y < 0. \end{cases}}$$

Each layer has uniform $\bar{\omega}_z$, and the base flow is steady and parallel: $D(\nabla^2 \bar{\psi})/Dt = \bar{U} \partial_x (\nabla^2 \bar{\psi}) = 0$, satisfying (8).

(c) Linearised mode

Total $\psi = \bar{\psi}(y) + \text{Re}[\hat{\psi}(y)e^{ikx-i\omega t}]$. Linearise (8):

$$\left(\frac{\partial}{\partial t} + \bar{U}\frac{\partial}{\partial x}\right) \nabla^2 \psi' - (\nabla^2 \bar{\psi})_y \frac{\partial \psi'}{\partial x} = 0.$$

Within each layer $\bar{U}'' = 0$, so:

$$(-i\omega + ik\bar{U})(-k^2\hat{\psi} + \hat{\psi}'') = 0.$$

Cancel $\omega - k\bar{U} \neq 0$:

$$\boxed{\hat{\psi}'' - k^2\hat{\psi} = 0} \quad (9)$$

Solutions decaying as $|y| \rightarrow \infty$ (take $k > 0$):

$$\hat{\psi}(y) = \begin{cases} C_+ e^{-ky}, & y > 0, \\ C_- e^{+ky}, & y < 0. \end{cases}$$

Pressure from horizontal momentum (linearised, $u'_x = \partial_y \psi'$, $u'_y = -\partial_x \psi'$):

$$\frac{\partial u'_x}{\partial t} + \bar{U} \frac{\partial u'_x}{\partial x} + u'_y \frac{\partial \bar{U}}{\partial y} = -\frac{1}{\rho} \frac{\partial p'}{\partial x}.$$

Substitute mode form $u'_x = \hat{\psi}' e^{i(kx-\omega t)}$, $u'_y = -ik\hat{\psi} e^{i(kx-\omega t)}$:

$$(-i\omega + ik\bar{U})\hat{\psi}' + (-ik\hat{\psi})\bar{U}' = -\frac{ik}{\rho}\hat{p}.$$

Factor i on the left:

$$i(k\bar{U} - \omega)\hat{\psi}' - ik\hat{\psi}\bar{U}' = -\frac{ik}{\rho}\hat{p}.$$

Divide both sides by $-ik$:

$$\frac{\omega - k\bar{U}}{k}\hat{\psi}' + \hat{\psi}\bar{U}' = \frac{\hat{p}}{\rho}.$$

Solve for $\hat{p}$:

$$\boxed{\hat{p} = \rho \left[\frac{\omega - k\bar{U}}{k} \frac{\partial \hat{\psi}}{\partial y} + \hat{\psi} \frac{\partial \bar{U}}{\partial y} \right]} \quad (10)$$

Interface matching at $y = \eta$. Continuity of normal velocity / displacement gives $[\hat{\psi}]_-^+ = 0$ at $y = 0$. Continuity of pressure (no surface tension) gives, using (10) at $\bar{U}(0) = 0$:

$$\left[\frac{\omega}{k} \frac{\partial \hat{\psi}}{\partial y} + \hat{\psi} \frac{\partial \bar{U}}{\partial y} \right]_-^+ = 0 \quad \text{at } y = 0.$$

Apply continuity of $\hat{\psi}$ at $y = 0$:

$$C_+ = C_- \equiv C.$$

Differentiate the layer solutions:

$$\hat{\psi}'(0^+) = -kC, \quad \hat{\psi}'(0^-) = +kC.$$

Read off base shear from $\bar{U}(y) = -\alpha y$ ($y > 0$), $\bar{U}(y) = 0$ ($y < 0$):

$$\bar{U}'(0^+) = -\alpha, \quad \bar{U}'(0^-) = 0.$$

Insert into the second matching condition:

$$\left[\frac{\omega}{k}(-kC) + C(-\alpha) \right] - \left[\frac{\omega}{k}(+kC) + C(0) \right] = 0.$$

Expand:

$$-\omega C - \alpha C - \omega C = 0.$$

Collect:

$$-2\omega C - \alpha C = 0.$$

Factor $C \neq 0$ and solve for ω :

$$\boxed{\omega = -\frac{\alpha}{2}} \quad (11)$$

Streamfunction (to within multiplicative constant):

$$\boxed{\hat{\psi}(y) = C e^{-k|y|}}$$

Evolution. ω is real $\Rightarrow$ disturbance is neutrally stable: amplitude constant in time. Phase factor $e^{ikx-i\omega t} = e^{ik(x+\alpha t/(2k))}$, giving phase speed $c = \omega/k = -\alpha/(2k)$ in $\hat{\mathbf{x}}$ (wave travels in $-x$ for $\alpha > 0$). Spatial structure: $\hat{\psi} \propto e^{-k|y|}$, decays exponentially away from the interface — a localised vorticity-edge wave on the kink in $\bar{U}(y)$.

Question 4 — Lubrication: rotating cylinder near a plane

Problem. Long rigid cylinder, axis $\parallel \hat{\mathbf{y}}$, radius a , rotating $\Omega \hat{\mathbf{y}}$ above plane $z = 0$, immersed in incompressible viscous fluid.

(a) Reynolds number

$$\boxed{\text{Re} = \frac{UL}{\nu} = \frac{\rho UL}{\mu}}$$

ratio of inertial to viscous forces. High $\text{Re} \gg 1$: inertia dominates, viscous effects confined to thin boundary layers and wakes; turbulence possible. Low $\text{Re} \ll 1$ (Stokes flow): viscous balance, time-reversible, flow $\nabla^2 \mathbf{u} = \nabla p / \mu$ linear in boundary forcing, no separation, no inertia.

(b)(i) Geometry, BCs, surface velocity

Cylinder axis at $z = a + h_0$. Surface: $z_{\text{cyl}} = a + h_0 - \sqrt{a^2 - x^2}$. Gap height:

$$h(x) = z_{\text{cyl}}(x) = h_0 + a - \sqrt{a^2 - x^2}.$$

Binomial expand for $x \ll a$:

$$\sqrt{a^2 - x^2} = a \sqrt{1 - x^2/a^2} \approx a - \frac{x^2}{2a}.$$

Substitute:

$$\boxed{h(x) \approx h_0 + \frac{x^2}{2a}} \quad (12)$$

Boundary conditions. No-slip + impermeability on rigid surfaces:

$$\mathbf{u}(z=0) = \mathbf{0}, \quad \mathbf{u}|_{\text{cyl surface}} = \Omega \hat{\mathbf{y}} \times \mathbf{r}_{\text{from axis}}.$$

Position of contact-region surface point relative to axis:

$$\mathbf{r}_{\text{from axis}} = x \hat{\mathbf{x}} - (\sqrt{a^2 - x^2}) \hat{\mathbf{z}}.$$

Apply $\hat{\mathbf{y}} \times \hat{\mathbf{x}} = -\hat{\mathbf{z}}$, $\hat{\mathbf{y}} \times \hat{\mathbf{z}} = \hat{\mathbf{x}}$:

$$\Omega \hat{\mathbf{y}} \times \mathbf{r}_{\text{from axis}} = \Omega x (-\hat{\mathbf{z}}) - \Omega \sqrt{a^2 - x^2} \hat{\mathbf{x}}.$$

Collect:

$$\Omega \hat{\mathbf{y}} \times \mathbf{r}_{\text{from axis}} = -\Omega \sqrt{a^2 - x^2} \hat{\mathbf{x}} - \Omega x \hat{\mathbf{z}}.$$

Apply $x \ll a$ in $\sqrt{a^2 - x^2} \approx a$:

$$\Omega \hat{\mathbf{y}} \times \mathbf{r}_{\text{from axis}} \approx -\Omega a \hat{\mathbf{x}} - \Omega x \hat{\mathbf{z}}.$$

Drop $\hat{\mathbf{z}}$ -component as $O(\Omega x) \ll |\Omega a|$:

$$\boxed{\mathbf{u}(x, z = h(x)) \approx U \hat{\mathbf{x}}, \quad U = -\Omega a}$$

(b)(ii) Lubrication equation

Translation invariance: cylinder is long, $\partial_y \equiv 0$, $u_y \approx 0$. Aspect ratio $\epsilon \equiv h_0/a \ll 1$. From $\nabla \cdot \mathbf{u} = 0$:

$$u_z \sim \epsilon u_x.$$

Compare derivatives:

$$\frac{\partial^2 u_x}{\partial x^2} \sim u_x/a^2, \quad \frac{\partial^2 u_x}{\partial z^2} \sim u_x/h_0^2, \quad \text{ratio} = \epsilon^2 \ll 1.$$

Drop $\partial_x^2 u_x$. Reduced Reynolds number:

$$\frac{\rho u_x \partial_x u_x}{\mu \partial_z^2 u_x} \sim \frac{\rho U^2/a}{\mu U/h_0^2} = \frac{\rho U h_0^2}{\mu a} = \text{Re } \epsilon^2 \ll 1.$$

Drop inertia. Required: $h \ll a$ and $\rho U h^2/(\mu a) \ll 1$. The x -momentum equation reduces to

$$\boxed{\mu \frac{\partial^2 u_x}{\partial z^2} \approx -\frac{\partial p}{\partial x}} \quad (13)$$

Neglect $\partial_x p$ as instructed:

$$\frac{\partial^2 u_x}{\partial z^2} = 0.$$

Integrate twice in z :

$$u_x(x, z) = B(x)z + C(x).$$

Apply $u_x(z=0) = 0$:

$$C(x) = 0.$$

Apply $u_x(z=h) = U = -\Omega a$:

$$B(x) = \frac{U}{h(x)} = -\frac{\Omega a}{h(x)}.$$

Therefore

$$\boxed{u_x(x, z) \approx \frac{U z}{h(x)} = -\frac{\Omega a z}{h(x)}} \quad (14)$$

(linear Couette across the gap, shear rate $U/h(x)$).

(c) Force and torque

Viscous shear stress in the fluid:

$$\sigma_{xz}|_{z=h} = \mu \frac{\partial u_x}{\partial z} \Big|_{z=h} = \frac{\mu U}{h(x)} = -\frac{\mu \Omega a}{h(x)}.$$

Traction on cylinder from fluid (outward normal from cylinder is $-\hat{\mathbf{z}}$ at the bottom):

$$t_x = -\sigma_{xz}|_{z=h} = \frac{\mu \Omega a}{h(x)}.$$

Net horizontal force on the cylinder (length L , integrated over the thin-gap strip $|x| < b$):

$$F \approx \int_{-b}^b \frac{\mu \Omega a}{h(x)} L dx = \int_{-b}^b \frac{A}{h} dx, \quad A = \mu \Omega a L.$$

Torque about the axis (lever arm $\mathbf{r} = -a\hat{\mathbf{z}}$, force $d\mathbf{F} = t_x L dx \hat{\mathbf{x}}$):

$$d\mathbf{\Gamma} = (-a\hat{\mathbf{z}}) \times (t_x L dx) \hat{\mathbf{x}} = -a t_x L dx (\hat{\mathbf{z}} \times \hat{\mathbf{x}}).$$

Apply $\hat{\mathbf{z}} \times \hat{\mathbf{x}} = \hat{\mathbf{y}}$:

$$\Gamma \approx - \int_{-b}^b \frac{\mu \Omega a^2 L}{h(x)} dx = \int_{-b}^b \frac{B'}{h} dx, \quad B' = -\mu \Omega a^2 L.$$

Evaluate the integral with (12) extended to $b \rightarrow \infty$:

$$\int_{-\infty}^{\infty} \frac{dx}{h_0 + x^2/(2a)}.$$

Substitute $x = \sqrt{2ah_0} \tan \theta$:

$$dx = \sqrt{2ah_0} \sec^2 \theta d\theta, \quad h = h_0 \sec^2 \theta.$$

Insert into the integral:

$$\int_{-\infty}^{\infty} \frac{dx}{h(x)} = \frac{\sqrt{2ah_0}}{h_0} \int_{-\pi/2}^{\pi/2} d\theta.$$

Evaluate:

$$\int_{-\infty}^{\infty} \frac{dx}{h(x)} = \sqrt{\frac{2a}{h_0}} \pi.$$

Therefore

$$\boxed{F \approx \pi \mu \Omega a L \sqrt{\frac{2a}{h_0}}, \quad \Gamma \approx -\pi \mu \Omega a^2 L \sqrt{\frac{2a}{h_0}}} \quad (15)$$

$|F|$ and $|\Gamma|$ both diverge as $h_0 \rightarrow 0$ — characteristic lubrication amplification. Signs: $F > 0$ (fluid drag opposes the $-\hat{\mathbf{x}}$ surface motion at the bottom), $\Gamma < 0$ (viscous torque opposes rotation).

(d) Free roll-and-slide

If the cylinder also translates with $V(t)\hat{\mathbf{x}}$, the surface point at the bottom moves at

$$\mathbf{v}_{\text{surf}} = V\hat{\mathbf{x}} + \Omega\hat{\mathbf{y}} \times (-a\hat{\mathbf{z}}).$$

Apply $\hat{\mathbf{y}} \times \hat{\mathbf{z}} = \hat{\mathbf{x}}$:

$$\mathbf{v}_{\text{surf}} = (V - \Omega a)\hat{\mathbf{x}}.$$

The plate is fixed: top BC becomes $u_x(z = h) = V - \Omega a$. The lubrication analysis is linear in U , so replace $-\Omega a \rightarrow V - \Omega a$, equivalently $\Omega a \rightarrow \Omega a - V$, in (c):

$$F \approx \pi\mu L \sqrt{\frac{2a}{h_0}} (\Omega a - V),$$

$$\Gamma \approx -\pi\mu L a \sqrt{\frac{2a}{h_0}} (\Omega a - V).$$

Define

$$\lambda \equiv \pi\mu L \sqrt{2a/h_0}.$$

Equations of motion (slowly varying, no fluid inertia):

$$M \frac{dV}{dt} = F = \lambda(\Omega a - V), \quad I \frac{d\Omega}{dt} = \Gamma = -\lambda a(\Omega a - V).$$

Track the slip $s \equiv \Omega a - V$:

$$\frac{ds}{dt} = a \frac{d\Omega}{dt} - \frac{dV}{dt}.$$

Substitute equations of motion:

$$\frac{ds}{dt} = -\frac{\lambda a^2}{I} s - \frac{\lambda}{M} s.$$

Combine:

$$\frac{ds}{dt} = -\frac{1}{\tau} s, \quad \frac{1}{\tau} = \lambda \left(\frac{a^2}{I} + \frac{1}{M} \right).$$

Integrate:

$$s(t) = s(0) e^{-t/\tau} \xrightarrow[t \rightarrow \infty]{} 0.$$

Therefore

$$\boxed{V(t \rightarrow \infty) = \Omega(t \rightarrow \infty) a}$$

Pure rolling. The combination $MV + I\Omega/a$ is conserved by the dynamics: $\frac{d}{dt} = F + \Gamma/a = \lambda(\Omega a - V) - \lambda(\Omega a - V) = 0$.